

JEE Mains 2019 Chapter wise Question Bank

Differential Equation – Questions

Q1

If $y = y(x)$ is the solution of the differential equation,

$x \frac{dy}{dx} + 2y = x^2$ satisfying $y(1) = 1$, then $y\left(\frac{1}{2}\right)$ is equal to:

- (1) $\frac{7}{64}$ (2) $\frac{1}{4}$
(3) $\frac{49}{16}$ (4) $\frac{13}{16}$

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Q2

Let $f: [0, 1] \rightarrow \mathbb{R}$ be such that $f(xy) = f(x) \cdot f(y)$, for all $x, y \in [0, 1]$, and $f(0) \neq 0$. If $y = y(x)$ satisfies the differential equation, $\frac{dy}{dx} = f(x)$ with $y(0) = 1$, then

$y\left(\frac{1}{4}\right) + y\left(\frac{3}{4}\right)$ is equal to:

- (1) 3 (2) 4
(3) 2 (4) 5

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Q3

If $\frac{dy}{dx} + \frac{3}{\cos^2 x} y = \frac{1}{\cos^2 x}$, $x \in \left(-\frac{\pi}{3}, \frac{\pi}{3}\right)$, and

$y\left(\frac{\pi}{4}\right) = \frac{4}{3}$, then $y\left(-\frac{\pi}{4}\right)$ equals:

- (1) $\frac{1}{3} + e^6$ (2) $\frac{1}{3}$
(3) $-\frac{4}{3}$ (4) $\frac{1}{3} + e^3$

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Q4

The curve amongst the family of curves represented by the differential equation, $(x^2 - y^2) dx + 2xy dy = 0$ which passes through $(1, 1)$, is:

- (1) a circle with centre on the x -axis.
(2) an ellipse with major axis along the y -axis.
(3) a circle with centre on the y -axis.
(4) a hyperbola with transverse axis along the x -axis.

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Q5

If $y(x)$ is the solution of the differential equation

$\frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x}$, $x > 0$, where $y(1) = \frac{1}{2}e^{-2}$, then :

- (1) $y(\log_e 2) = \log_e 4$
(2) $y(\log_e 2) = \frac{\log_e 2}{4}$
(3) $y(x)$ is decreasing in $\left(\frac{1}{2}, 1\right)$
(4) $y(x)$ is decreasing in $(0, 1)$

Differential Equation

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Q6

The solution of the differential equation, $\frac{dy}{dx} = (x-y)^2$,

when $y(1) = 1$, is :

(1) $\log_e \left| \frac{2-x}{2-y} \right| = x-y$

(2) $-\log_e \left| \frac{1-x+y}{1+x-y} \right| = 2(x-1)$

(3) $-\log_e \left| \frac{1+x-y}{1-x+y} \right| = x+y-2$

(4) $\log_e \left| \frac{2-y}{2-x} \right| = 2(y-1)$

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Q7

Let $y = y(x)$ be the solution of the differential equation,

$x \frac{dy}{dx} + y = x \log_e x$, ($x > 1$). If $2y(2) = \log_e 4 - 1$, then $y(e)$ is equal to :

(1) $-\frac{e}{2}$ (2) $-\frac{e^2}{2}$

(3) $\frac{e}{4}$ (4) $\frac{e^2}{4}$

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Q8

Let $y = y(x)$ be the solution of the differential equation, $(x^2 + 1)^2 \frac{dy}{dx} + 2x(x^2 + 1)y = 1$ such that $y(0) = 0$. If

$\sqrt{a} y(1) = \frac{\pi}{32}$, then the value of 'a' is :

(1) $\frac{1}{4}$ (2) $\frac{1}{2}$ (3) 1 (4) $\frac{1}{16}$

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Q9

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The solution of the differential equation $x \frac{dy}{dx} + 2y = x^2$ ($x \neq 0$) with $y(1) = 1$, is:

(1) $y = \frac{4}{5}x^3 + \frac{1}{5x^2}$ (2) $y = \frac{x^3}{5} + \frac{1}{5x^2}$
(3) $y = \frac{x^2}{4} + \frac{3}{4x^2}$ (4) $y = \frac{3}{4}x^2 + \frac{1}{4x^2}$

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Q10

If $\cos x \frac{dy}{dx} - y \sin x = 6x$, ($0 < x < \frac{\pi}{2}$) and $y\left(\frac{\pi}{3}\right) = 0$, then $y\left(\frac{\pi}{6}\right)$ is equal to:

(1) $\frac{\pi^2}{2\sqrt{3}}$ (2) $-\frac{\pi^2}{2}$ (3) $-\frac{\pi^2}{2\sqrt{3}}$ (4) $-\frac{\pi^2}{4\sqrt{3}}$

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Q11

If $y = y(x)$ is the solution of the differential equation $\frac{dy}{dx} = (\tan x - y) \sec^2 x$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, such that $y(0) = 0$,

then $y\left(-\frac{\pi}{4}\right)$ is equal to :

(1) $e - 2$ (2) $\frac{1}{2} - e$ (3) $2 + \frac{1}{e}$ (4) $\frac{1}{e} - 2$

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Q12

Let $y = y(x)$ be the solution of the differential equation,

$$\frac{dy}{dx} + y \tan x = 2x + x^2 \tan x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \text{ such that } y(0) = 1.$$

Then :

$$(1) \quad y\left(\frac{\pi}{4}\right) + y\left(-\frac{\pi}{4}\right) = \frac{\pi^2}{2} + 2$$

$$(2) \quad y\left(\frac{\pi}{4}\right) + y\left(-\frac{\pi}{4}\right) = -\sqrt{2}$$

$$(3) \quad y\left(\frac{\pi}{4}\right) - y\left(-\frac{\pi}{4}\right) = \sqrt{2}$$

$$(4) \quad y'\left(\frac{\pi}{4}\right) - y'\left(-\frac{\pi}{4}\right) = \pi - \sqrt{2}$$

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Q13

Consider the differential equation, $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$. If

value of y is 1 when $x = 1$, then the value of x for which $y = 2$, is :

$$(1) \quad \frac{5}{2} + \frac{1}{\sqrt{e}} \quad (2) \quad \frac{3}{2} - \frac{1}{\sqrt{e}} \quad (3) \quad \frac{1}{2} + \frac{1}{\sqrt{e}} \quad (4) \quad \frac{3}{2} - \sqrt{e}$$

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Q14

The general solution of the differential equation

$$(y^2 - x^3) dx - xy dy = 0 \quad (x \neq 0) \text{ is :}$$

$$(1) \quad y^2 - 2x^2 + cx^3 = 0 \quad (2) \quad y^2 + 2x^3 + cx^2 = 0$$

$$(3) \quad y^2 + 2x^2 + cx^3 = 0 \quad (4) \quad y^2 - 2x^3 + cx^2 = 0$$

(where c is a constant of integration)

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Differential Equation - Solutions

Q1

(3) Since, $x \frac{dy}{dx} + 2y = x^2$

$$\Rightarrow \frac{dy}{dx} + \frac{2}{x}y = x$$

I.F. $= e^{\int \frac{2}{x} dx} = e^{2 \ln x} = e^{\ln x^2} = x^2.$

Solution of differential equation is:

$$y \cdot x^2 = \int x \cdot x^2 dx$$

$$y \cdot x^2 = \frac{x^4}{4} + C$$

$$\therefore y(1) = 1$$

$$\therefore C = \frac{3}{4}$$

Then, from equation (1)

$$y \cdot x^2 = \frac{x^4}{4} + \frac{3}{4}$$

$$\therefore y = \frac{x^2}{4} + \frac{3}{4x^2}$$

$$\therefore y\left(\frac{1}{2}\right) = \frac{1}{16} + 3 = \frac{49}{16}$$

$$(1) f(xy) = f(x)f(y) \quad \dots(1)$$

Put $x = y = 0$ in (1) to get $f(0) = 1$

Put $x = y = 1$ in (1) to get $f(1) = 0$ or $f(1) = 1$

$f(1) = 0$ is rejected else $y = 1$ in (1) gives $f(x) = 0$

imply $f(0) = 0$.

Hence, $f(0) = 1$ and $f(1) = 1$

By first principle derivative formula,

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} f(x) \left(\frac{f\left(1 + \frac{h}{x}\right) - f(1)}{h} \right) \end{aligned}$$

$$\Rightarrow f'(x) = \frac{f(x)}{x} f'(1)$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \frac{k}{x} \Rightarrow \ln f(x) = k \ln x + c$$

$$f(1) = 1 \Rightarrow \ln 1 = k \ln 1 + c \Rightarrow c = 0$$

$$\Rightarrow \ln f(x) = k \ln x \Rightarrow f(x) = x^k \text{ but } f(0) = 1$$

$$\Rightarrow k = 0$$

$$\therefore f(x) = 1$$

$$\frac{dy}{dx} = f(x) = 1 \Rightarrow y = x + c, y(0) = 1 \Rightarrow c = 1$$

$$\Rightarrow y = x + 1$$

$$\therefore y\left(\frac{1}{4}\right) + y\left(\frac{3}{4}\right) = \frac{1}{4} + 1 + \frac{3}{4} + 1 = 3$$

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Q2

Q3

Differential Equation

(1) Given, $\frac{dy}{dx} + \frac{3}{\cos^2 x} y = \frac{1}{\cos^2 x}$

$$\frac{dy}{dx} = \sec^2 x (1 - 3y)$$

$$\Rightarrow \int \frac{dy}{(1-3y)} = \int \sec^2 x \, dx$$

$$\Rightarrow -\frac{1}{3} \ln |1-3y| = \tan x + C \quad \dots(i)$$

$$\because y\left(\frac{\pi}{4}\right) = \frac{4}{3} \quad (\text{Given})$$

$$\Rightarrow -\frac{1}{3} \ln |1-4| = \tan \frac{\pi}{4} + C$$

$$\Rightarrow -\frac{1}{3} \ln 3 = C + 1 \Rightarrow C = -1 - \frac{1}{3} \ln 3$$

$\therefore$ in eq. (i), we get

$$-\frac{1}{3} \ln |1-3y| = \tan x - 1 - \frac{1}{3} \ln 3$$

$$\text{Put, } x = -\frac{\pi}{4}$$

$$\begin{aligned} \Rightarrow -\frac{1}{3} \ln |1-3y| &= \tan\left(-\frac{\pi}{4}\right) - 1 - \frac{1}{3} \ln 3 \\ &= -1 - 1 - \frac{1}{3} \ln 3 \end{aligned}$$

$$\Rightarrow \ln |1-3y| = 6 + \ln 3$$

$$\Rightarrow \ln \left| \frac{1}{3} - y \right| = 6 \Rightarrow \left| \frac{1}{3} - y \right| = e^6 \Rightarrow y = \frac{1}{3} \pm e^6$$

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Q4

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$$y^2 dx - 2xy dy = x^2 dx$$

$$2xy dy - y^2 dx = -x^2 dx$$

$$d(xy^2) = -x^2 dx$$

$$\frac{xd(y^2) - y^2 d(x)}{x^2} = -dx$$

$$d\left(\frac{y^2}{x}\right) = -dx$$

$$\int d\left(\frac{y^2}{x}\right) = -\int dx$$

$$\frac{y^2}{x} = -x + C \quad \dots(1)$$

Since, the above curve passes through the point (1, 1)

$$\text{Then, } \frac{1^2}{1} = -1 + C \Rightarrow C = 2$$

Now, the curve (1) becomes

$$y^2 = -x^2 + 2x$$

$$\Rightarrow y^2 = -(x-1)^2 + 1$$

$$(x-1)^2 + y^2 = 1$$

The above equation represents a circle with centre (1, 0) and centre lies on x-axis.

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Q5

Differential Equation

(3) Given differential equation is,

$$\frac{dy}{dx} + \left(2 + \frac{1}{x}\right)y = e^{-2x}, x > 0$$

$$\text{IF} = e^{\int \left(2 + \frac{1}{x}\right) dx} = e^{2x + \ln x} = xe^{2x}$$

Complete solution is given by

$$y(x) \cdot x e^{2x} = \int x e^{2x} \cdot e^{-2x} dx + c$$

$$= \int x dx + c$$

$$y(x) \cdot e^{2x} \cdot x = \frac{x^2}{2} + c$$

$$\text{Given, } y(1) = \frac{1}{2}e^{-2}$$

$$\therefore \frac{1}{2}e^{-2} \cdot e^2 \cdot 1 = \frac{1}{2} + c \Rightarrow c = 0$$

$$\therefore y(x) = \frac{x^2}{2} \cdot \frac{e^{-2x}}{x}$$

$$y(x) = \frac{x}{2} \cdot e^{-2x}$$

Differentiate both sides with respect to x ,

$$y'(x) = \frac{e^{-2x}}{2} (1 - 2x) < 0 \quad \forall x \in \left(\frac{1}{2}, 1\right)$$

Hence, $y(x)$ is decreasing in $\left(\frac{1}{2}, 1\right)$

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Q6

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(2) The given differential equation

$$\frac{dy}{dx} = (x - y)^2 \quad \dots(1)$$

$$\text{Let } x - y = t \Rightarrow 1 - \frac{dy}{dx} = \frac{dt}{dx}$$

$$\Rightarrow \frac{dy}{dx} = 1 - \frac{dt}{dx}$$

Now, from equation (1)

$$\left(1 - \frac{dt}{dx}\right) = (t)^2$$

$$\Rightarrow 1 - t^2 = \frac{dt}{dx} \Rightarrow \int dx = \int \frac{dt}{1 - t^2}$$

$$\Rightarrow -x = \frac{1}{2 \times 1} \ln \left| \frac{t-1}{t+1} \right| + c$$

$$\Rightarrow -x = \frac{1}{2} \ln \left| \frac{x-y-1}{x-y+1} \right| + c$$

$\therefore$ The given condition $y(1) = 1$

$$-1 = \frac{1}{2} \ln \left| \frac{1-1-1}{1-1+1} \right| + c \Rightarrow c = -1$$

$$\text{Hence, } 2(x-1) = -\ln \left| \frac{1-x+y}{1-y+x} \right|$$

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Q7

Differential Equation

(3) Consider the differential equation,

$$\frac{dy}{dx} + \frac{y}{x} = \log_e x$$

$$\therefore IF = e^{\int \frac{1}{x} dx} = x$$

$$\therefore yx = \int x \ln x \, dx$$

$$\Rightarrow xy = \ln x \cdot \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} dx$$

$$\Rightarrow xy = \frac{x^2}{2} \cdot \ln x - \frac{x^2}{4} + c$$

$$\text{Given, } 2y(2) = \log_e 4 - 1.$$

$$\therefore 2y = 2 \ln 2 - 1 + c$$

$$\Rightarrow \ln 4 - 1 = \ln 4 - 1 + c$$

$$\text{i.e. } c = 0$$

$$\Rightarrow xy = \frac{x^2}{2} \ln x - \frac{x^2}{4}$$

$$\Rightarrow y = \frac{x}{2} \ln x - \frac{x}{4}$$

$$\Rightarrow y(e) = \frac{e}{2} - \frac{e}{4} = \frac{e}{4}$$

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Q8

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$$(4) (1+x^2)^2 \frac{dy}{dx} + 2x(1+x^2)y = 1$$

$$\Rightarrow \frac{dy}{dx} + \left(\frac{2x}{1+x^2} \right) y = \frac{1}{(1+x^2)^2}$$

Since, the above differential equation is a linear differential equation

$$\therefore I.F. = e^{\int \frac{2x}{1+x^2} dx} = e^{\log(1+x^2)} = 1+x^2$$

Then, the solution of the differential equation

$$\Rightarrow y(1+x^2) = \int \frac{dx}{1+x^2} + c$$

$$\Rightarrow y(1+x^2) = \tan^{-1} x + c \quad \dots(1)$$

If $x = 0$ then $y = 0$ (given)

$$\Rightarrow 0 = 0 + c$$

$$\Rightarrow c = 0$$

Then, equation (1) becomes,

$$\Rightarrow y(1+x^2) = \tan^{-1} x$$

Now put $x = 1$ in above equation, then

$$2y = \frac{\pi}{4}$$

$$\Rightarrow 2\left(\frac{\pi}{32\sqrt{a}}\right) = \frac{\pi}{4} \quad \left[\sqrt{a}y(1) = \frac{\pi}{32} \right]$$

$$\Rightarrow \sqrt{a} = \frac{1}{4}$$

$$\Rightarrow a = \frac{1}{16}$$

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Q9

$$(3) \frac{dy}{dx} + \frac{2}{x}y = x \text{ and } y(1) = 1 \text{ (given)}$$

Since, the above differential equation is the linear

differential equation, then $I.F. = e^{\int \frac{2}{x} dx} = x^2$

Now, the solution of the linear differential equation

$$y \times x^2 = \int x^3 dx$$

$$\Rightarrow yx^2 = \frac{x^4}{4} + C$$

$$\therefore y(1) = 1$$

$$\therefore 1 \times 1 = \frac{1}{4} + C \Rightarrow C = \frac{3}{4}$$

$\therefore$ solution becomes

$$y = \frac{x^2}{4} + \frac{3}{4x^2}$$

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Differential Equation

Q10

$$(3) \cos x dy - (\sin x) y dx = 6x dx \\ \Rightarrow \int d(y \cos x) = \int 6x dx \Rightarrow y \cos x = 3x^2 + C \dots (1)$$

$$\text{Given, } y\left(\frac{\pi}{3}\right) = 0$$

Putting $x = \frac{\pi}{3}$ and $y = 0$ in eq. (1), we get

$$(10) \times \left(\frac{1}{2}\right) = \frac{3\pi^2}{9} + C \Rightarrow C = \frac{-\pi^2}{3}$$

$$\text{So, from (1) } y \cos x = 3x^2 - \frac{\pi^2}{3}$$

Now, put — in the above equation,

$$y \frac{\sqrt{3}}{2} = \frac{3\pi^2}{36} - \frac{\pi^2}{3} \Rightarrow \frac{\sqrt{3}y}{2} = \frac{-3\pi^2}{12} \Rightarrow y = \frac{-\pi^2}{2\sqrt{3}}$$

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Q11

$$(1) \frac{dy}{dx} + y \sec^2 x = \sec^2 x \times \tan x$$

Given equation is linear differential equation.

$$\text{IF} = e^{\int \sec^2 x dx} = e^{\tan x}$$

$$\Rightarrow y \cdot e^{\tan x} = \int e^{\tan x} \sec^2 x \times \tan x dx$$

$$\text{Put } \tan x = u = \sec^2 x dx = du$$

$$y e^{\tan x} = \int e^u u du \Rightarrow y e^{\tan x} = u e^u - e^u + c$$

$$\Rightarrow y e^{\tan x} = (\tan x - 1) e^{\tan x} + c$$

$$\Rightarrow y = (\tan x - 1) + c \cdot e^{-\tan x}$$

$$\therefore y(0) = 0 \text{ (given)} \Rightarrow 0 = -1 + c \Rightarrow c = 1$$

Hence, solution of differential equation,

$$y\left(-\frac{\pi}{4}\right) = -1 - 1 + e = -2 + e$$

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Q12

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(4) Given differential equation is,

$$\frac{dy}{dx} + y \tan x = 2x + x^2 \tan x$$

Here, $P = \tan x$, $Q = 2x + x^2 \tan x$

$$\text{I.F.} = e^{\int \tan x dx} = e^{\ln|\sec x|} = |\sec x|$$

$$\therefore y(\sec x) = \int (2x + x^2 \tan x) \sec x dx$$

$$= \int x^2 \tan x \sec x dx + \int 2x \sec x dx = x^2 \sec x + c$$

$$\text{Given } y(0) = 1 \Rightarrow c = 1$$

$$\therefore y = x^2 + \cos x \dots (i)$$

Now put $x = \frac{\pi}{4}$ and $x = -\frac{\pi}{4}$ in equation (i),

$$y\left(\frac{\pi}{4}\right) = \frac{\pi^2}{16} + \frac{1}{\sqrt{2}} \text{ and } y\left(-\frac{\pi}{4}\right) = \frac{\pi^2}{16} + \frac{1}{\sqrt{2}}$$

$$\Rightarrow y\left(\frac{\pi}{4}\right) - y\left(-\frac{\pi}{4}\right) = 0$$

$$\frac{dy}{dx} = 2x - \sin x$$

$$\therefore y'\left(\frac{\pi}{4}\right) = \frac{\pi}{2} - \frac{1}{\sqrt{2}} \text{ and } y'\left(-\frac{\pi}{4}\right) = -\frac{\pi}{2} + \frac{1}{\sqrt{2}}$$

$$\Rightarrow y'\left(\frac{\pi}{4}\right) - y'\left(-\frac{\pi}{4}\right) = \pi - \sqrt{2}$$

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Q13

(2) Consider the differential equation,

$$y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$$

$$\Rightarrow \frac{dx}{dy} + \left(\frac{1}{y^2}\right)x = \frac{1}{y^3}$$

$$\text{I.F.} = e^{\int \frac{1}{y^2} dy} = e^{-\frac{1}{y}}$$

$$\therefore x \cdot e^{-\frac{1}{y}} = \int e^{-\frac{1}{y}} \cdot \frac{1}{y^3} dy + c$$

$$\text{Put } -\frac{1}{y} = u \Rightarrow \frac{1}{y^2} dy = du$$

$$\Rightarrow x \cdot e^{-\frac{1}{y}} = -\int u e^u du + c = -u e^u + e^u + c$$

$$\Rightarrow x \cdot e^{-\frac{1}{y}} = e^{-\frac{1}{y}} \left(\frac{1}{y} + 1\right) + c$$

$$\text{At } y = 1, x = 1$$

$$1 = 2 + ce \Rightarrow c = -\frac{1}{e} \Rightarrow x = \left(1 + \frac{1}{y}\right) - \frac{1}{e} e^{\frac{1}{y}}$$

$$\text{On putting } y = 2, \text{ we get } x = \frac{3}{2} - \frac{1}{\sqrt{e}}$$

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Q14

(2) Given differential equation can be written as,

$$y^2 dx - xy dy = x^3 dx$$

$$\Rightarrow \frac{(y dx - x dy) y}{x^2} = x dx \Rightarrow -y d\left(\frac{y}{x}\right) = x dx$$

$$\Rightarrow -\frac{y}{x} \cdot d\left(\frac{y}{x}\right) = dx \Rightarrow -\frac{1}{2} \left(\frac{y}{x}\right)^2 = x + c_1$$

$$\Rightarrow 2x^3 + cx^2 + y^2 = 0 \quad [\text{Here, } c = 2c_1]$$

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